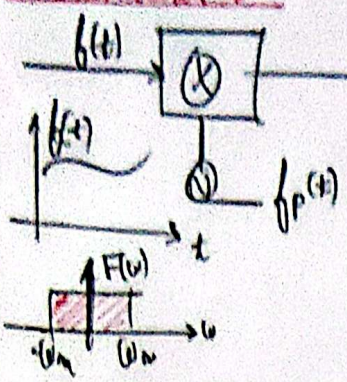
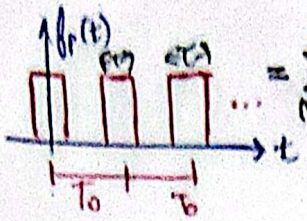
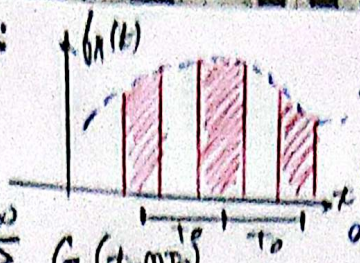
PAM Natural:



Tendremos:



$$= \sum_{n=-\infty}^{\infty} G_\tau(t-nT_0)$$



⇒ ¿Cómo la Señal Original

Ahora la Señal de Muestreo es un Tren de Pulsos Periódicos (Más Realizable)

⇒ τ: Ancho del Pulso y $\frac{1}{T_0}$: f_0 ⇒ Frecuencia de Muestreo

⇒ **Hallamos $F_M(\omega)$:** $f_M(t) = f(t) \times f_p(t) \Rightarrow F_M(\omega) = \frac{1}{2\pi}\left[F(\omega) * F_p(\omega)\right] = \frac{1}{2\pi}\left[F(\omega) * \mathcal{F}\left(\sum_{n=-\infty}^{\infty} G_\tau(t)\right)\right]$

⇒ **Hallamos: $F_p(\omega)$:**

$$F_p(\omega) = 2\pi \sum_{n=-\infty}^{\infty} F_n \cdot \delta(\omega - n\omega_0)$$

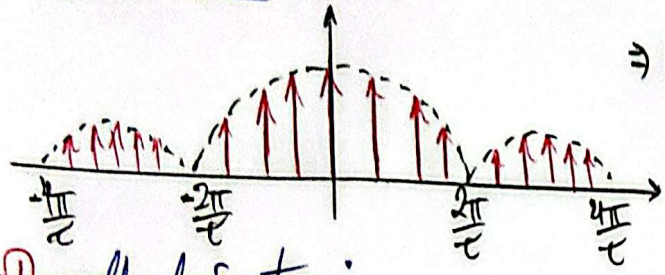
⇒ Donde $F_n = \frac{1}{T_0}\int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} G_\tau(t) e^{-jn\omega_0 t}\, dt$

$$F_n = \frac{1}{T_0}\left(\tau\, Sa\left(\frac{n\omega_0 \tau}{2}\right)\right) = \frac{\tau}{T_0} Sa\left(\frac{n\omega_0\tau}{2}\right)$$

$\omega_0 = \frac{2\pi}{T_0}$

⇒ **Entonces:** $F_p(\omega) = 2\pi \sum_{n=-\infty}^{\infty} \frac{\tau}{T_0} Sa\left(\frac{n\omega_0\tau}{2}\right) \cdot \delta(\omega - n\omega_0)$

⇒: **Gráfica de $F_p(\omega)$**



$$\Rightarrow F_M(\omega) = \frac{1}{2\pi}\left[F(\omega) * 2\pi \sum_{n=-\infty}^{\infty} \frac{\tau}{T_0} Sa\left(\frac{n\omega_0\tau}{2}\right)\delta(\omega - n\omega_0)\right]$$

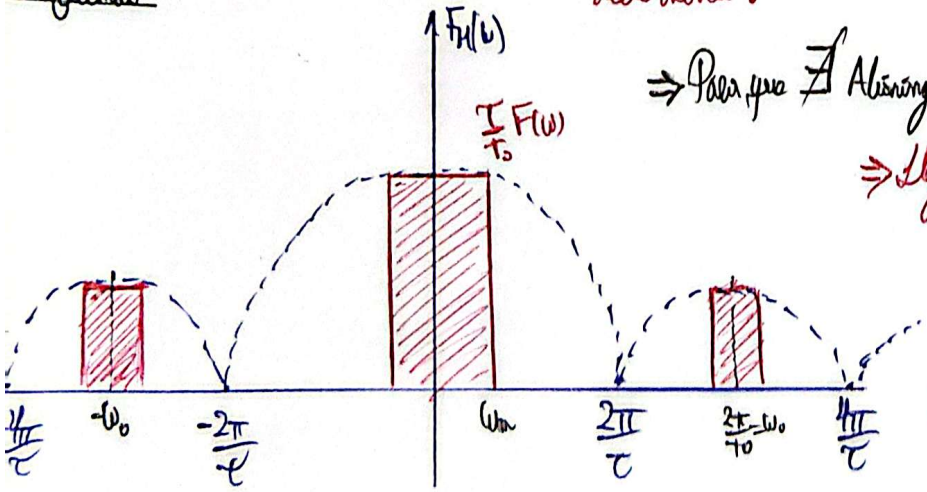
$$\boxed{F_M(\omega) = \frac{\tau}{T_0}\sum_{n=-\infty}^{\infty} Sa\left(\frac{n\omega_0\tau}{2}\right) F(\omega - n\omega_0)}$$

⇒ **Desarrollando Sumatoria:**

$$F_M(\omega) = \underbrace{\frac{\tau}{T_0} Sa(0)}_{\text{Número}} F(\omega) + \underbrace{\frac{\tau}{T_0} Sa\left(\frac{\omega_0\tau}{2}\right)}_{\text{Número}} \cdot F(\omega - \omega_0) + \frac{\tau}{T_0} Sa\left(\frac{2\omega_0\tau}{2}\right) F(\omega - 2\omega_0) + \cdots$$

⇒ "Solo modifican la proporción de la Señal más no la distorsiona."

Graficando:



⇒ Para que ∄ Aliasing ⇒ $\omega_0 - \omega_m \geq \omega_m$

$\omega_0 \geq 2\omega_m$ Nyquist

⇒ Llegamos a un análisis similar de muestreo

¿Alguna restricción para τ?

obviamente $\tau < T_0$

$$\boxed{\text{Duty Cycle} = \frac{\tau}{T_0}\ (0,1,\ldots 0,9)}$$

Escaneado con CamScanner